

Platforms for Growth: Multinationals, Trade, and Technology Diffusion

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EIIT 2026

Introduction

- **Export platform:** strong multinational production (MP) presence + export orientation
 - Maquiladoras in Northern Mexico
 - Ireland's pharmaceutical sector
 - Special Economic Zones throughout China and Southeast Asia

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- **What are the effects of becoming an export platform on growth?**
 - + Attract highly productive multinational firms
 - – Competition for domestic firms, stifles domestic innovation
- Challenging question to answer because...
 - Endogeneity of export orientation
 - Endogeneity of openness to foreign firms
 - General equilibrium effects + dynamics

This Paper

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- **Empirics**
 - Exogenous trade policy reform (BTA) + instrument for openness to foreign firms
 - BTA effects on exports nearly $2\times$ as large in high-foreign-share sectors
 - Foreign firms more responsive + domestic firms enjoy positive spillovers

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- **Vietnam**: fast-growing developing country with important role for export platforms
- **Empirics**
 - Exogenous trade policy reform (BTA) + instrument for openness to foreign firms
 - BTA effects on exports nearly $2\times$ as large in high-foreign-share sectors
 - Foreign firms more responsive + domestic firms enjoy positive spillovers
- **Quantitative theory**
 - Model of trade + multinational production + technology diffusion
 - BTA raised Vietnam's real wage by 1.65%: $4\times$ the gain without MP mobility or learning
 - Gain rested on prior openness: a Vietnam closed to multinationals gains only 0.21%

Related Literature

- **Trade shocks on exports, firms, and employment** (De Loecker and Goldberg '14, Dix-Carneiro and Kovak '16, McCaig and Pavcnik '18, Alfaro-Ureña, Manelici and Vasquez '22, Kovak and Morrow '22, McCaig, Pavcnik and Wong '25)
 - ⇒ Empirics + theory quantify how effects of trade vary with foreign firm presence
- **Idea diffusion and growth** (Sampson '16, Buera and Oberfield '20, Perla, Tonetti and Waugh '21, Benhabib, Perla and Tonetti '22, Cai and Xiang '25)
 - ⇒ Identify role of learning from exogenous shifts in source distribution of ideas
- **Multinational production spillovers** (Aitken and Harrison '99, Javorcik '04, Keller and Yeaple '09, Poole '13, Alfaro '16, Alfaro and Chen '18, Setzler and Tintelnot '21, Alfaro-Ureña, Manelici and Vasquez '22)
 - ⇒ Stronger spillovers in export oriented sectors

Roadmap

Empirical Evidence

Model

Quantification

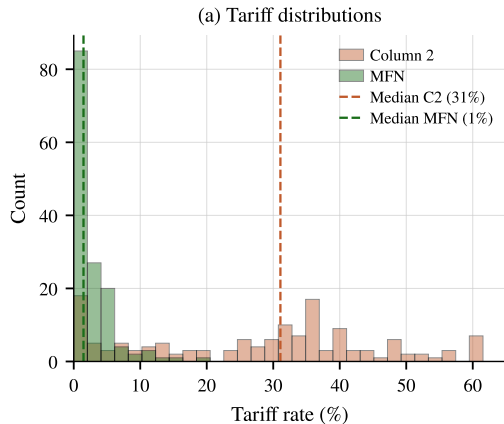
Counterfactuals

Data

- Vietnam Enterprise Survey (GSO), 2000–2015
 - Covers all registered enterprises; 580,000+ firm-year obs in manufacturing
 - Ownership: foreign, domestic private, state
- Technology and Competitiveness Survey (VTC), 2010–2014
 - R&D, technology transfer, MP connections for 7,900 firms
- Orbis, 1996–2019
 - Global multinational firm data
- UN Comtrade/BACI: bilateral trade flows (1996–2015)

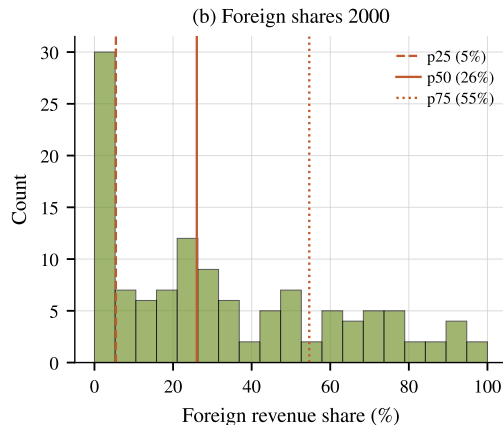
The 2001 US-Vietnam Bilateral Trade Agreement (BTA)

- US cut tariffs on Vietnamese goods
 - From Column 2 to MFN tariff schedule
 - Average tariff: 32% $\rightarrow$ 3%
 - Immediate reduction in Dec 2001
- Neither US nor Vietnamese industries could negotiate sector-specific cuts
- No changes in Vietnamese import tariffs
- Share of US in Vietnamese exports:
 - 2000 $\Rightarrow$ 5%
 - 2003 $\Rightarrow$ 24%



Multinational Presence Pre-BTA

- Vietnam was open to foreign firms before BTA
- F_i = share of foreign firms in total revenue in year 2000 in sector i
 - Foreign = $\geq 50\%$ foreign equity
 - Median = 26%
 - p25 = 5%, p75 = 55%
- Japan, Taiwan, South Korea, Singapore account for $\sim 2/3$ of all multinational revenues



Do BTA Effects Depend on Initial Foreign Presence?

- Main regression specification

$$\Delta \log Y_{it} = \alpha_t + \beta T_i + \gamma F_i + \delta T_i F_i + X'_{it} \eta + \epsilon_{it}$$

- 4-digit manufacturing sectors i , years $t = 2002, \dots, 2015$, changes relative to 2000
- $T_i = \log(1 + \tau_i^{\text{Col2}}) - \log(1 + \tau_i^{\text{MFN}})$
- F_i = share of foreign firms in sector i 's revenue in 2000
- T_i and F_i demeaned $\implies \beta$ = BTA effect at the mean foreign share
- Sectors weighted by 2000 exports; SEs clustered by sector
- X_{it} : Chinese exports to US, RoW tariffs on VN, textile quota $\times$ year

Origin Country Specialization as an IV

- Example: SIC code 3430 = automobile parts
 - 3.5% of all MP revenues in Japan, 1% in South Korea vs. 0.3% overall
 - Vietnam is “exposed” to Japan and South Korea $\implies$ high foreign share in this sector
 - Data: 48% of all revenues in this sector in Vietnam from foreign firms, vs. 26% median

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- Instrument definition

S_{io} = share of sector i in MP revenues in origin country o

$\hat{S}_{io} = S_{io} - [\text{mean of } S_{io} \text{ across origins } o]$

ω_o = share of origin country o in MP revenues in Vietnam

$$Z_i = \sum_o \omega_o \hat{S}_{io}$$

- Instrument for F_i and $T_i F_i$ using Z_i and $T_i Z_i$

No Correlation with T_i

IV Illustration

BTA → Vietnam's Exports to US

$$\Delta \log Y_{it} = \alpha_t + \beta T_i + \gamma F_i + \delta T_i F_i + X'_{it} \eta + \epsilon_{it}$$

	Full Sample (2002-2015)	Pre BTA
	OLS	
T_i	9.64*** (2.26)	
F_i		
$T_i \times F_i$		
Controls	X	
Observations	1,083	
KP Wald F	–	

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	Full Sample (2002-2015)				Pre BTA
	OLS	OLS	IV	IV	
T_i	9.64*** (2.26)	10.58*** (1.90)	11.38*** (1.52)	8.69*** (2.41)	
F_i		3.32*** (1.22)	6.25*** (0.57)	6.49*** (0.94)	
$T_i \times F_i$		20.05** (8.46)	37.42*** (7.28)	28.23*** (5.99)	
Controls	✗	✗	✗	✓	
Observations	1,083	1,083	1,083	1,083	
KP Wald F	–	–	41.44	24.80	

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	OLS	OLS	IV	IV	1996-99
T_i	9.64*** (2.26)	10.58*** (1.90)	11.38*** (1.52)	8.69*** (2.41)	-1.14 (1.61)
F_i		3.32*** (1.22)	6.25*** (0.57)	6.49*** (0.94)	0.02 (0.91)
$T_i \times F_i$		20.05** (8.46)	37.42*** (7.28)	28.23*** (5.99)	-0.20 (5.84)
Controls	✗	✗	✗	✓	✓
Observations	1,083	1,083	1,083	1,083	255
KP Wald F	–	–	41.44	24.80	18.51

Composition or Spillovers?

- **Composition**: foreign firms are more responsive to trade shocks
⇒ Larger average effect of BTA on foreign firm employment + revenue

Composition or Spillovers?

- **Composition**: foreign firms are more responsive to trade shocks
⇒ Larger average effect of BTA on foreign firm employment + revenue
- **Spillovers**: in sectors with a high foreign share, all firms respond strongly
⇒ Positive interaction effect within domestic (and foreign?) firms

Composition or Spillovers?

$$\Delta \log Y_{it} = \alpha_t + \beta T_i + \gamma F_i + \delta T_i F_i + X'_{it} \eta + \epsilon_{it}$$

	Foreign Firms		Domestic Firms	
	Employment	Revenue	Employment	Revenue
T_i	3.35*** (1.15)	3.09*** (1.15)	0.94 (1.24)	1.78* (1.06)
F_i	-0.27 (0.55)	-1.63*** (0.56)	1.93*** (0.62)	0.74* (0.39)
$T_i \times F_i$	6.68** (2.82)	-1.15 (2.91)	9.92** (3.86)	6.13** (2.77)
Controls	✓	✓	✓	✓
Observations	1,045	1,044	1,113	1,113
F-statistic	24.33	24.33	25.26	25.26

Domestic Firms Increase R&D and Technology Adoption

$$Y_{it} = \alpha_t + \beta T_i + \gamma F_i + \delta T_i F_i + X'_{it} \eta + \epsilon_{it}$$

	# MNC Connected	Tech Transfer	R&D	Modify Technology	# Completed Modifications
T_i	1.31 (0.81)	0.05 (0.07)	0.04 (0.14)	0.11*** (0.03)	0.42*** (0.12)
F_i	0.92** (0.38)	0.11*** (0.04)	0.52*** (0.12)	0.03 (0.02)	0.10 (0.07)
$T_i \times F_i$	12.81** (5.09)	1.98*** (0.38)	3.05*** (1.06)	0.94*** (0.22)	2.45*** (0.61)
Controls	✓	✓	✓	✓	✓
Observations	320	320	396	396	396
KP Wald F	20.30	20.30	20.84	20.84	20.84

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Model Outline

- Building blocks
 - Multisector trade model with input-output linkages (Caliendo Parro 2015)
 - Multinational production (Ramondo Rodríguez-Clare 2013)
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 - MPs can choose to produce in any location
 - Extra margin of adjustment $\implies$ more responsive to trade cost shocks

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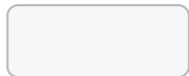
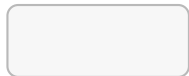
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- **Learning from export platforms**
 - MPs who export overcome (i) trade costs and (ii) MP costs
 - Those who do both successfully are highly selected $\implies$ highly productive
 - Highly productive $\implies$ large spillovers to domestic firms via learning

Environment

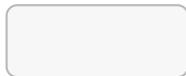
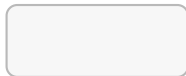
- **N countries**
 - Buyers i , producers j , headquarters h
- **S sectors**
 - Each sector consists of a continuum of CES varieties $\omega \in [0, 1]$
 - Varieties are consumed directly and used as an input in production
 - Many perfectly competitive firms can produce $\omega \implies$ lowest-cost producer wins
- **Continuous time**
 - Ideas diffuse gradually $\implies$ productivity distribution evolves over time

Model in Pictures

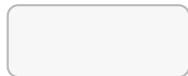
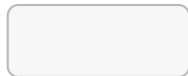
Headquarters (h)



Production locations (j)

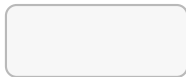


Destinations (i)



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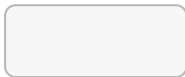
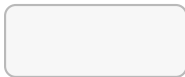
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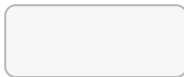
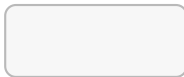
λ_h^s



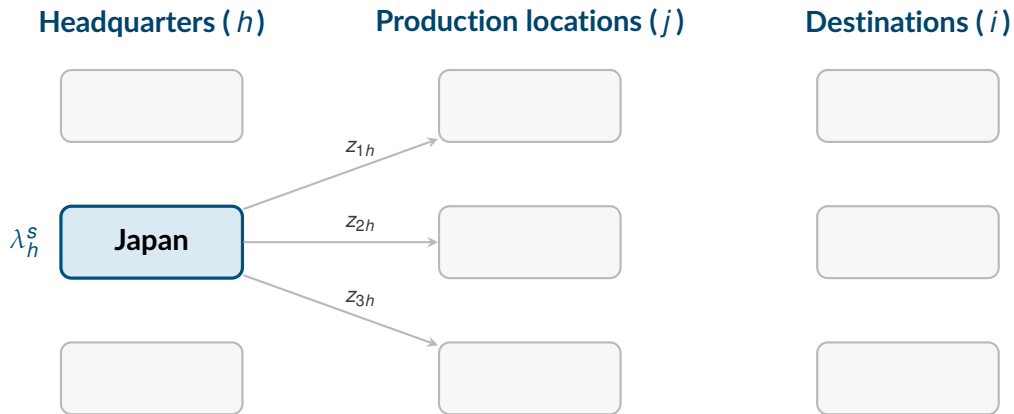
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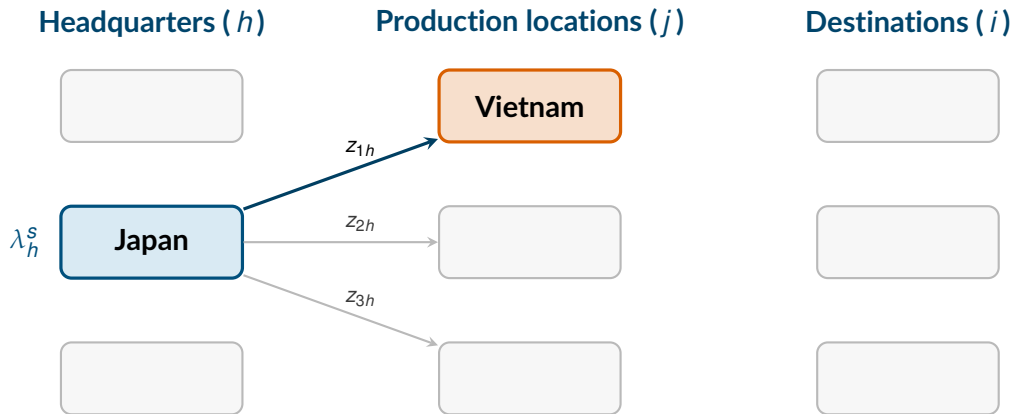
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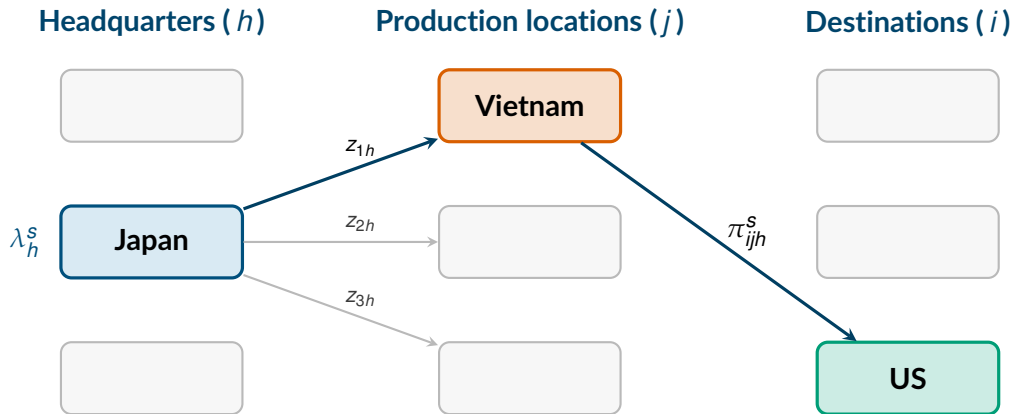
Model in Pictures



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Firms

- Each headquarters-sector pair has many identical potential producers of variety ω
- Characterized by **vector of productivities in different production locations j**

$$z_h^s(\omega) = [z_{1h}^s(\omega), z_{2h}^s(\omega), \dots, z_{Nh}^s(\omega)]$$

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- z_h^s is drawn from a **multivariate Fréchet distribution**
- Scale = λ_h^s = technology level
- Shape = θ = idiosyncratic productivity dispersion across varieties
- Correlation = η = substitutability across different production locations

Production + Trade

- Produce with Cobb-Douglas bundle of labor + intermediates $\implies$ input costs x_j^s
- Cost of headquarters h to produce one unit in j and ship to i

$$c_{ijh}^s(\omega) = \frac{\overbrace{x_j^s}^{\text{input costs}} \cdot \overbrace{\delta_{jh}^s}^{\text{MP cost}} \cdot \overbrace{\tau_{ijh}^s}^{\text{trade cost}}}{z_{jh}^s(\omega)} \equiv \frac{\kappa_{ijh}^s}{z_{jh}^s(\omega)}$$

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- Perfect competition + multivariate Fréchet $\implies$ prices and trade shares

Derivation

Technology Diffusion

- Vector of productivities z_h^s evolves over time
- Combine **new ideas** with **learning from firms producing in h**
 - Importance of learning controlled by “learning intensity” ρ
- Same structure as Buera Oberfield (2020), except:
 - Many different HQ locations produce in each location h
 - Sample firms in proportion to employment shares

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 - Many different HQ locations produce in each location h
 - Sample firms in proportion to employment shares
- Result: law of motion for λ

$$\frac{d\lambda_h^s}{dt} = \tilde{\alpha}_h^s \sum_{i,k} \ell_{ihk}^s (\tilde{\lambda}_{ihk}^s)^\rho, \quad \tilde{\lambda}_{ihk}^s = \frac{\lambda_k^s}{\pi_{ihk}^s (\psi_{ihk}^s)^{-\eta}}$$

Derivation

Two Forces Behind Export-Platform Effects

Partial equilibrium: wages and prices fixed; small trade cost cut $T^s \equiv -\Delta \log \tau_{ijh}^s$ from j to i

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- **Multinational mobility** ($\rho = 0$)

$$\Delta \log \pi_{ijj}^s = \theta T^s \quad \text{vs.} \quad \Delta \log \pi_{ijh}^s = \frac{\theta}{1 - \eta} T^s \quad (h \neq j)$$

$\implies$ direct BTA effect larger for foreign than domestic firms, but *no* spillover to domestic firms

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- **Learning from export platforms** ($\eta = 0$), long run:

$$\tilde{\lambda}_j^s \propto \left(\frac{\tilde{\lambda}_j^s}{\pi_{jjj}^s} \right)^\rho \times \sum_i \sum_h \ell_{ijh}^s \left(\delta_{jh}^s \tau_{ijh}^s \frac{P_i^s}{P_j^s} \right)^{\rho\theta}$$

$\implies$ exporting MNEs pay δ and τ : doubly selected, best ideas $\implies$ spillovers

Roadmap

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Calibration Outline

- **Baseline** = year 2000 (pre-BTA balanced growth path) Baseline
 - 4 countries: VN, USA, EA (JPN + KOR + TWN + SGP), RoW
 - 58 ISIC Rev. 3 sectors
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- **Hat algebra** (Dekle Eaton Kortum 2007) Hat Algebra
 - Difference out trade costs, multinational production costs, productivities...
 - Welfare: $\widehat{W}_j = \widehat{w}_j / \prod_s (\widehat{P}_j^s)^{\alpha_j^s}$, BTA path vs. no-BTA path, long run

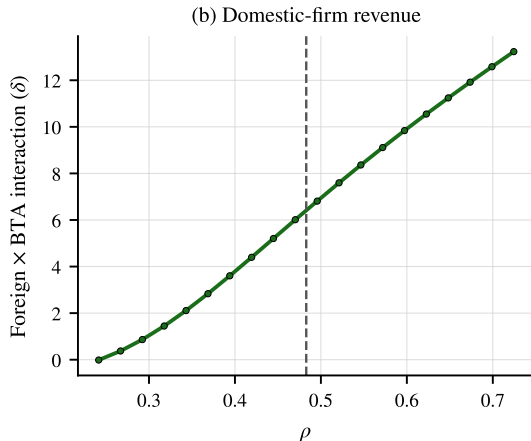
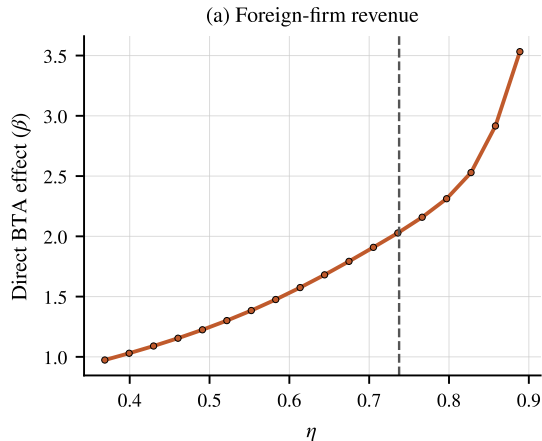
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- **Parameters**: estimate (θ, η, ρ) by indirect inference
 - BTA + interaction effects on exports to US
 - BTA + interaction effects on foreign revenue + employment
 - BTA + interaction effects on domestic revenue + employment

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 - BTA + interaction effects on exports to US
 - BTA + interaction effects on foreign revenue + employment
 - BTA + interaction effects on domestic revenue + employment
- **Result**: $\theta = 7.26$ (trade elasticities 4–8), $\eta = 0.74$ (Arkolakis et al. '18: 0.55), $\rho = 0.48$ (Buera–Oberfield '20: 0.60; Cai–Xiang '25: 0.73)

Identification in Pictures



- Direct BTA effect on **foreign revenue** rises in η
- Foreign $\times$ BTA interaction on **domestic revenue** rises in ρ

Roadmap

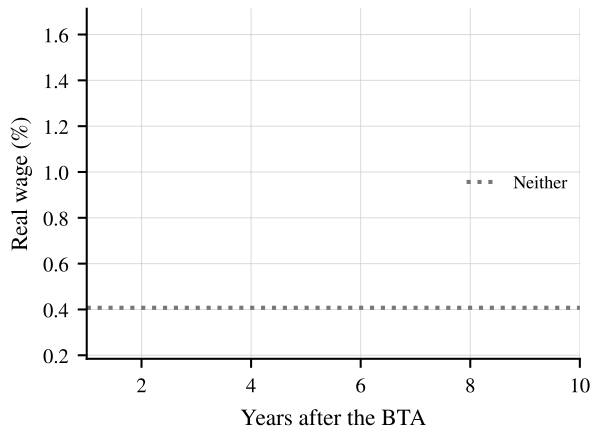
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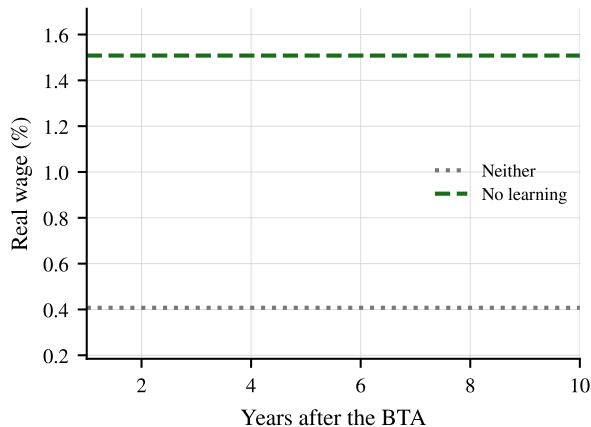
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Gains from the BTA Are Large and Driven by Multinationals



Long-run gain, %		
	Mobility off	Mobility on
Learning off	0.41	
Learning on		

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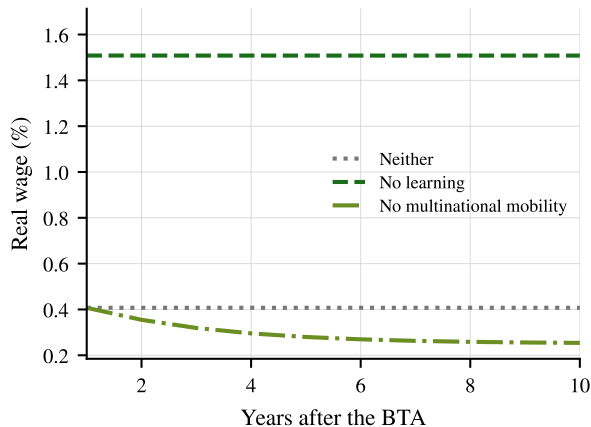


Real wage path relative to no-BTA path

	Long-run gain, %	
	Mobility off	Mobility on
Learning off	0.41	1.51
Learning on		

- MPs relocate: elasticity $\theta / (1 - \eta) = 27.7$ vs. $\theta = 7.3$

Gains from the BTA Are Large and Driven by Multinationals

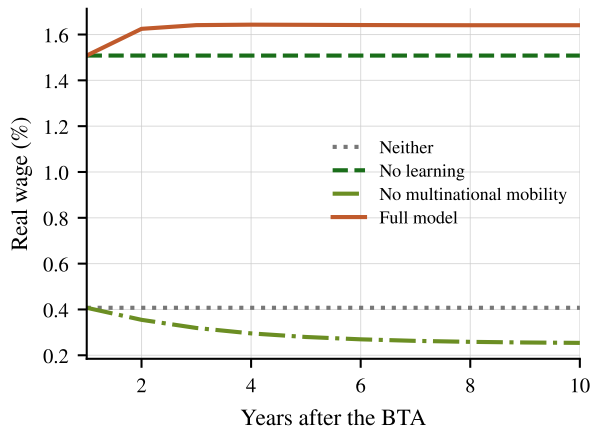


Real wage path relative to no-BTA path

	Long-run gain, %	
	Mobility off	Mobility on
Learning off	0.41	1.51
Learning on	0.25	

- Cheaper exporting $\Rightarrow$ less selected exporters $\Rightarrow$ worse ideas
- No MP inflow to offset

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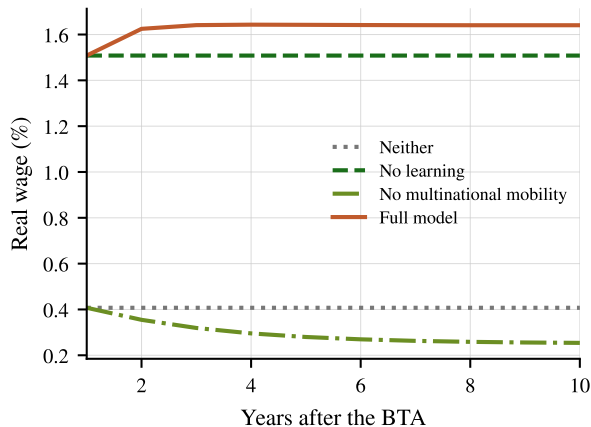


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- MPs move in with better ideas
- Channels are complements

Gains from the BTA Are Large and Driven by Multinationals



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	Mobility off	Mobility on
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Learning on	0.25	1.65

- $4\times$ the gain with neither channel
- 1.62% within two years

The BTA Gain Rested on Prior Openness to Multinationals

- Consider lowering pre-BTA foreign share; rerun the BTA
 - 1995 openness: foreign share of manufacturing output at 60% of its 2000 level

	Foreign share of tradable production	BTA gain, %	Without learning, %	Learning, pp
Observed, 2000	0.34	1.65	1.51	0.14

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Closed to multinationals	0	0.21	0.52	-0.32

- Early stage of opening: learning worth $3\times$ as much (27% of the gain vs. 8%)
- Closed: gain falls to 1/8

Conclusion

Empirics

- ⇒ Tariff cuts have much larger effects in sectors with strong foreign presence
- ⇒ Driven by both composition and spillovers to domestic firms
- ⇒ Evidence that domestic firms learn from multinationals

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- ⇒ Combine quantitative trade + multinational production + technology diffusion
- ⇒ Multinationals can relocate ⇒ more responsive
- ⇒ Exporting multinationals are highly selected ⇒ larger spillovers

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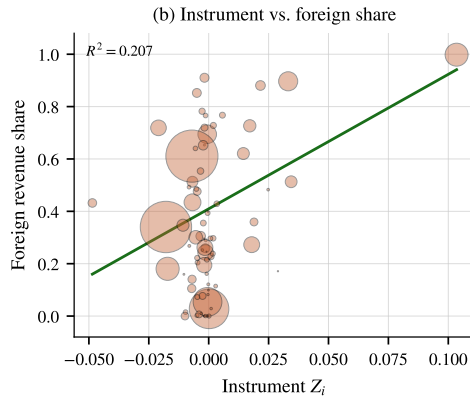
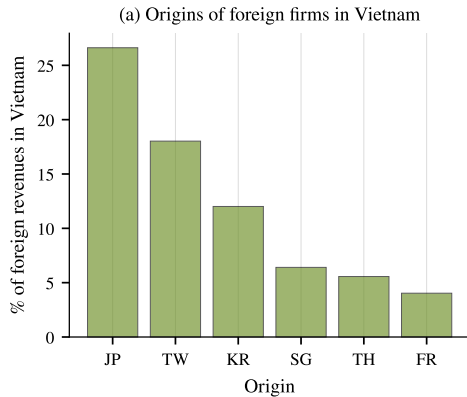
Quantification

- ⇒ BTA raised Vietnam's real wage by 1.65%; 0.41% without mobility or learning
- ⇒ Market access and openness to multinationals are worth far more together
- ⇒ Learning matters most at an early stage of opening

Data Sources

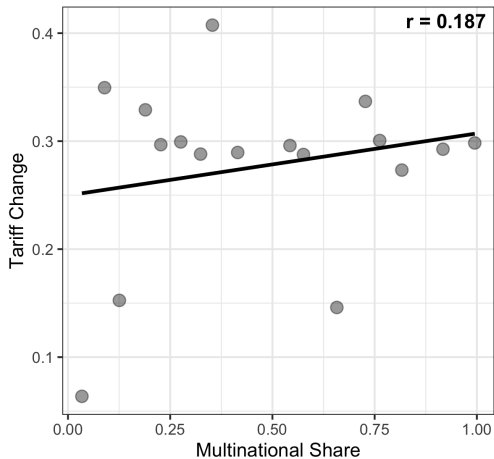
- Vietnam Enterprise Survey (GSO), 2000–2015
 - Census of registered enterprises; $\sim 580,000$ firm-year obs and 150,000 firms in manufacturing
 - Ownership: foreign ($\geq 50\%$ foreign equity), domestic private, state
- Technology and Competitiveness Survey (VTC), 2010–2014
 - R&D, technology transfer, MP connections for $\sim 7,900$ manufacturing firms
- Orbis, 1996–2019: global multinational firm data (instrument)
- USITC: US imports from Vietnam (1996–2015)
- UN Comtrade/BACI: bilateral trade flows (1996–2015)

IV Strategy Illustration



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Tariff Cuts Are Weakly Correlated with Initial Foreign Presence



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VN to US Exports: Post-BTA Subperiods

$$\Delta \log Y_{it} = \alpha_t + \beta T_i + \gamma F_i + \delta T_i F_i + X'_{it} \eta + \epsilon_{it}$$

	Full	Post BTA	
	2002-15	2002-06	2007-15
BTA	8.69*** (2.41)	6.75** (2.67)	9.75*** (2.45)
Foreign Share	6.49*** (0.94)	6.72*** (1.19)	6.47*** (0.90)
Foreign Share $\times$ BTA	28.23*** (5.99)	28.53*** (7.97)	30.50*** (7.00)
Controls	✓	✓	✓
Observations	1,083	385	698
KP Wald F	24.80	14.06	24.96

Composition or Spillovers? All Outcomes, Including Exporters

$$\Delta \log Y_{it} = \alpha_t + \beta T_i + \gamma F_i + \delta T_i F_i + X'_{it} \eta + \epsilon_{it}$$

	Domestic (Private) Firms			Foreign Firms		
	Exporters	Employment	Revenue	Exporters	Employment	Revenue
T_i	0.32 (0.46)	0.94 (1.24)	1.78* (1.06)	2.84*** (1.03)	3.35*** (1.15)	3.09*** (1.15)
F_i	-0.08 (0.36)	1.93*** (0.62)	0.74* (0.39)	0.18 (0.48)	-0.27 (0.55)	-1.63*** (0.56)
$T_i \times F_i$	4.41** (2.00)	9.92** (3.86)	6.13** (2.77)	3.82 (2.45)	6.68** (2.82)	-1.15 (2.91)
Controls	✓	✓	✓	✓	✓	✓
Observations	514	1,113	1,113	542	1,045	1,044
KP Wald F	4.39	25.26	25.26	22.64	24.33	24.33

Weak-Instrument Diagnostics (Lewis–Mertens 2026)

Specification	KP Wald F	Robust $g_{\min}$	$\tau = 0.10$		$\tau = 0.20$	
			Crit. value	Pass	Crit. value	Pass
<i>Exports to US</i>						
IV, no controls	41.44	16.03	28.36	✗	17.80	✗
IV, with controls [†]	24.80	28.43	27.82	✓	17.55	✓
Pre-BTA placebo, 1996–1999	18.51	7.89	31.06	✗	19.42	✗
<i>By ownership</i>						
Private: exporters	4.39	7.32	29.04	✗	18.26	✗
Private: employment [†]	25.26	29.84	29.13	✓	18.30	✓
Private: revenue [†]	25.26	29.84	28.10	✓	17.72	✓
Foreign: exporters	22.64	31.62	28.31	✓	17.82	✓
Foreign: employment [†]	24.33	29.74	28.55	✓	17.98	✓
Foreign: revenue [†]	24.33	29.73	28.19	✓	17.77	✓
<i>Technology outcomes (all five)</i>	20.30–20.84	21.97–22.61	24.10–24.54	✗	15.28–15.52	✓

Two endogenous regressors, so standard Stock–Yogo thresholds do not apply. Pass = worst-case 2SLS median bias below τ of benchmark at 5% level. [†] = calibration targets: all pass at $\tau = 0.10$. Anderson–Rubin sets exclude zero for the export platform effect.

Hat Algebra: Solving the Model in Changes

- Proportional changes (“hats”) in unit costs:

$$\hat{x}_j^s = \hat{w}_j^{\gamma_j^s} \left(\prod_{r=1}^S (\hat{P}_j^r)^{\beta_j^{sr}} \right)^{1-\gamma_j^s}$$

- Trade shares:

$$\hat{\pi}_{ijh}^s = \hat{\psi}_{ijh}^s \left(\frac{\hat{\lambda}_h^s (\hat{K}_{ih}^s)^{-\theta}}{\sum_{h'} \hat{\lambda}_{h'}^s (\hat{K}_{ih'}^s)^{-\theta}} \right)$$

- Technology dynamics:

$$\hat{\lambda}_h^s(t) = g_\lambda e^{(1-\rho)g_\lambda t} \sum_i \sum_k a_{ihk}^s \hat{\ell}_{ihk}^s(t) \left(\frac{\hat{\lambda}_k^s(t)}{\hat{\pi}_{ihk}^s(t) \hat{\psi}_{ihk}^s(t)^{-\eta}} \right)^\rho$$

a_{ihk}^s : baseline learning weight of (i, h, k) firms in country h

Baseline Trade Flow Construction

- Challenge: OECD reports flows by (producer, buyer); model needs (HQ, producer, buyer)
- Simplification: only EA multinational production, located in VN or EA
 - Trade costs $\tau_{ijh}^s = \tau_h^s \tau_{ij}^s$ for $i \neq j$
 - VES data: foreign vs. domestic firm revenues by sector, domestic vs. export sales
 - Pins down trilateral shares π_{ijh}^s for $j = \text{VN}$, $h \in \{\text{VN}, \text{EA}\}$
- Technology levels inferred from (symmetric trade-cost wedges, $\text{VN} = 1$):

$$\lambda_h^s \propto \left(\frac{X_h^s}{P_h^s} \right)^\theta \pi_{hhh}^s (\psi_{hhh}^s)^\eta$$

Related Literature

- **Trade shocks on exports, firms, and employment** (De Loecker and Goldberg '14, Dix-Carneiro and Kovak '16, McCaig and Pavcnik '18, Alfaro-Ureña, Manelici and Vasquez '22, Kovak and Morrow '22, McCaig, Pavcnik and Wong '25)
 - Interactions with foreign firm presence
- **Evidence on MPE spillovers** (Aitken and Harrison '99, Javorcik '04, Keller and Yeaple '09, Poole '13, Alfaro '16, Alfaro and Chen '18, Setzler and Tintelnot '21, Alfaro-Ureña, Manelici and Vasquez '22)
 - Identify the *amplifying* effects of foreign firms after trade liberalization
- **Theory on trade, MPE, and growth** (Ramondo and Rodríguez-Clare '13, Tintelnot '17, Arkolakis et al. '18, Sampson '16, Buera and Oberfield '20, Perla, Tonetti and Waugh '21, Benhabib, Perla and Tonetti '22, Cai and Xiang '25)
 - Model where MP mobility + technology diffusion jointly identified from IV estimates

Production Technology

- Roundabout production: firms use labor and composite intermediates from all sectors
- Output of a firm in sector s , HQ'd in h , producing in j :

$$y_{jh}^s(\omega) = \frac{z_{jh}^s(\omega)}{\delta_{jh}^s} f_j^s(l, m^1, \dots, m^S)$$

- f_j^s is Cobb-Douglas, common to all firms in (s, j)
- δ_{jh}^s : multinational production cost ($\delta_{hh}^s = 1$)
- Unit input cost bundle:

$$x_j^s = w_j^{\gamma_j^s} \left(\prod_{r=1}^S (P_j^r)^{\beta_j^{sr}} \right)^{1-\gamma_j^s}$$

- γ_j^s : labor share; β_j^{sr} : input share from sector r
- CES aggregation of varieties within each sector (elasticity σ)

Trade and Trilateral Shares

- Delivered cost for variety ω from (s, j, h) to destination i :

$$c_{ijh}^s(\omega) = \frac{\kappa_{ijh}^s}{z_{jh}^s(\omega)}, \quad \kappa_{ijh}^s \equiv x_j^s \delta_{jh}^s \tau_{ijh}^s$$

- Perfect competition: lowest-cost supplier wins each variety
- Trade shares and price indices (CES):

$$\pi_{ijh}^s = \int_{\Omega_{ijh}^s} \left(\frac{c_{ijh}^s(\omega)}{P_i^s} \right)^{1-\sigma} d\omega, \quad P_i^s = \left(\sum_j \sum_h \int_{\Omega_{ijh}^s} c_{ijh}^{s1-\sigma}(\omega) d\omega \right)^{\frac{1}{1-\sigma}}$$

- Production location shares:

$$\psi_{ijh}^s = \frac{\pi_{ijh}^s}{\sum_{j'} \pi_{ij'h}^s}$$

Share of h 's sales to i that are produced in j

Technology Diffusion: Setup

- Each firm draws a vector of potential productivities $z_h(\omega) = [z_{jh}(\omega)]_j$ from technological frontier G_h^s
- New ideas arrive as a Poisson process. Each idea decomposes as:

$$z = a \cdot b^\rho$$

- a : vector-valued *original component* (exogenous)
 - b : scalar *learning component* (endogenous)
 - $\rho \in [0, 1)$: learning intensity
- Arrival rate of ideas with original component exceeding a (Assumption 1):

$$A_h^s(a) = \alpha_h^s \left(\sum_j a_j^{-\theta/(1-\eta)} \right)^{1-\eta}$$

- α_h^s : exogenous arrival intensity
 - θ : shape of the right tail
 - η : correlation across locations in the productivity vector

Technology Diffusion: Source Distribution and Theorem

- “Learning from workers” (Buera & Oberfield, 2020): potential producers sample from all production technologies in use in their country

$$H_h^s(b) = \sum_i \sum_k \int_{\Omega_{ihk}^s} \ell_{ihk}^s(\omega) \mathbb{1}\{z_{hk}^s(\omega) \leq b\} d\omega$$

Sampling proportional to employment – larger firms contribute more

Theorem. The technological frontier G_h^s is multivariate Fréchet with shape θ , correlation η , and scale λ_h^s . The law of motion is:

$$\dot{\lambda}_h^s = \tilde{\alpha}_h^s \sum_k \sum_i \ell_{ihk}^s \left(\frac{\lambda_k^s}{\pi_{ihk}^s (\psi_{ihk}^s)^{-\eta}} \right)^\rho$$

where $\tilde{\alpha}_h^s = \alpha_h^s \Gamma(1 - \rho - \frac{\sigma-1}{\theta}) / \Gamma(1 - \frac{\sigma-1}{\theta})$ and ℓ_{ihk}^s is the employment share of (i, h, k) firms in country h .

Equilibrium

- Multivariate Fréchet yields closed-form trade shares:

$$\psi_{ijh}^s = \left(\frac{\kappa_{ijh}^s}{K_{ih}^s} \right)^{-\theta/(1-\eta)}, \quad K_{ih}^s \equiv \left(\sum_j \left(\kappa_{ijh}^s \right)^{-\frac{\theta}{1-\eta}} \right)^{-\frac{1-\eta}{\theta}}$$

$$\pi_{ijh}^s = \psi_{ijh}^s \cdot \frac{\lambda_h^s (K_{ih}^s)^{-\theta}}{\sum_{h'} \lambda_{h'}^s (K_{ih'}^s)^{-\theta}}$$

$$P_i^s \propto \left(\sum_h \lambda_h^s (K_{ih}^s)^{-\theta} \right)^{-1/\theta}$$

- Revenues and expenditures:

$$R_j^s = \sum_h \sum_i \pi_{ijh}^s X_i^s, \quad X_j^s = \alpha_j^s w_j l_j + \sum_r (1 - \gamma_j^r) \beta_j^{rs} R_j^r$$

- Labor market clearing: $w_j l_j = \sum_s \gamma_j^s R_j^s$

Balanced Growth and Partial Equilibrium

- α_h^s grows at rate $g_\alpha \implies$ all λ_h^s grow at rate $g_\lambda = g_\alpha / (1 - \rho)$
- Detrended technology on the balanced growth path:

$$\tilde{\lambda}_h^s = \frac{\tilde{\alpha}_h^s(0)}{g_\lambda} \sum_k \sum_i \ell_{ihk}^s \left(\frac{\tilde{\lambda}_k^s}{\pi_{ihk}^s (\psi_{ihk}^s)^{-\eta}} \right)^\rho$$

- Partial equilibrium: effect of a trade cost reduction $T^s \equiv -\Delta \log \tau_{ijh}^s$
 - Domestic firms: $\Delta \log \pi_{ijj}^s = \theta T^s$
 - Foreign firms: $\Delta \log \pi_{ijh}^s = \frac{\theta}{1 - \eta} T^s$
- Multinationals respond along an additional margin – choosing where to produce – with strength governed by η